

*CACS402: Cloud
computing
BCA 7th Semester*

Mechi Multiple Campus
Bhadrapur



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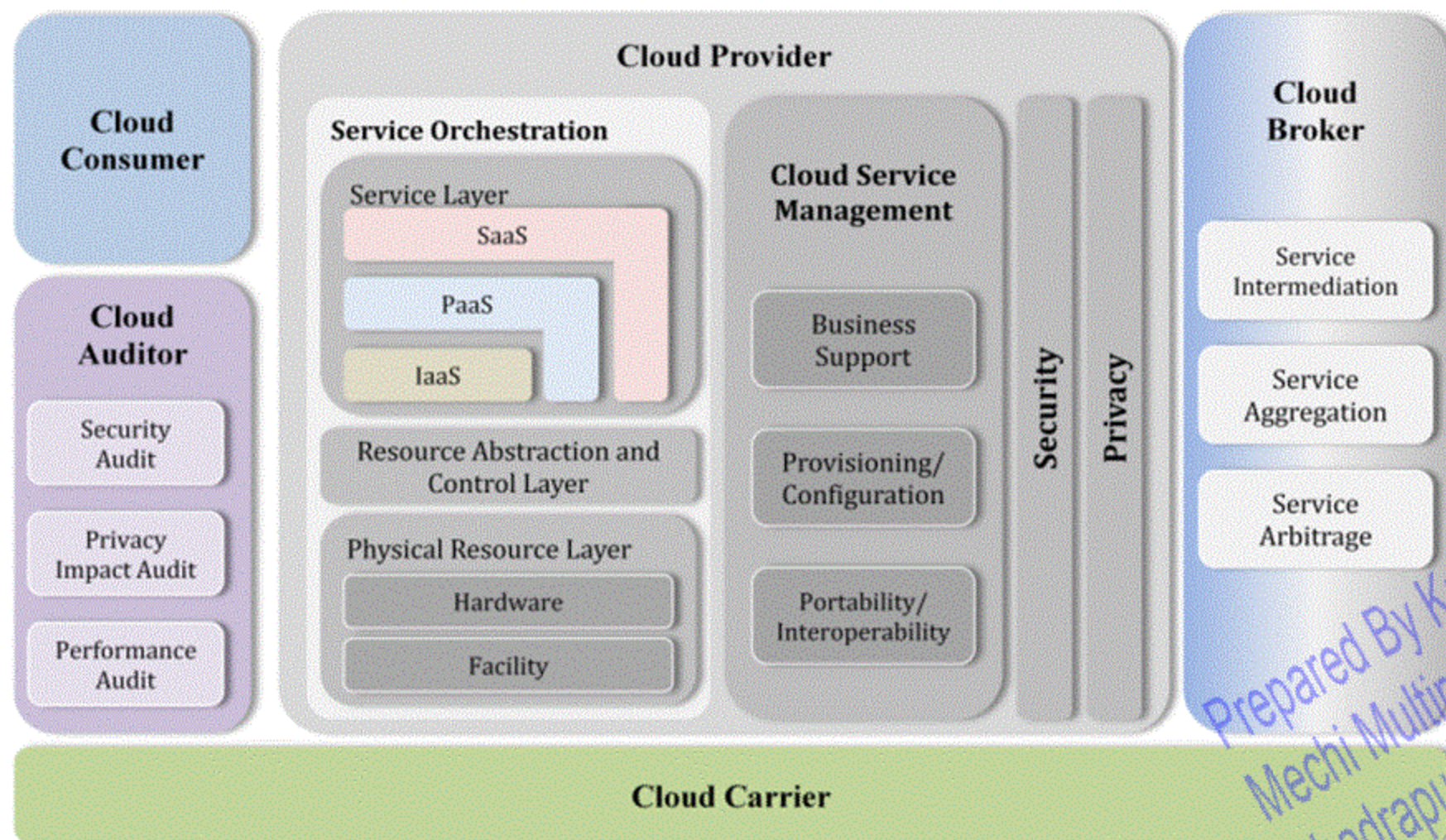
Unit 2: Cloud Computing Architecture

- » 2.1 Cloud reference model
 - » 2.1.1 Platform as service
 - » 2.1.2 Software as a service
 - » 2.1.3 Infrastructure as service
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- » 2.4 Security, trust and privacy

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NIST Cloud Computing Reference Architecture

(National Institute of Standards and Technology)



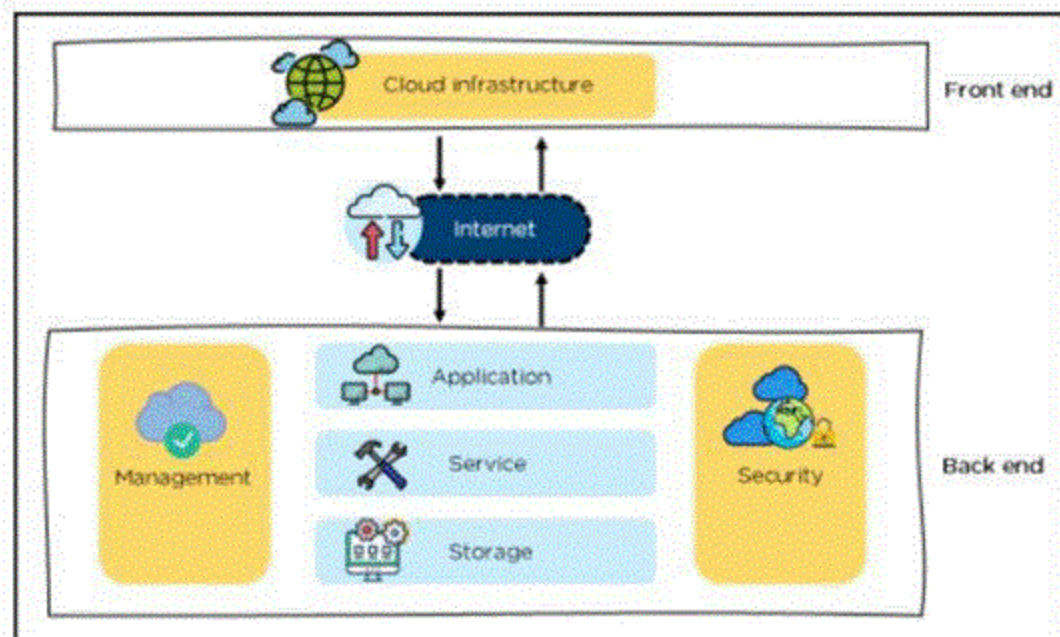
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NIST Cloud Computing Reference model

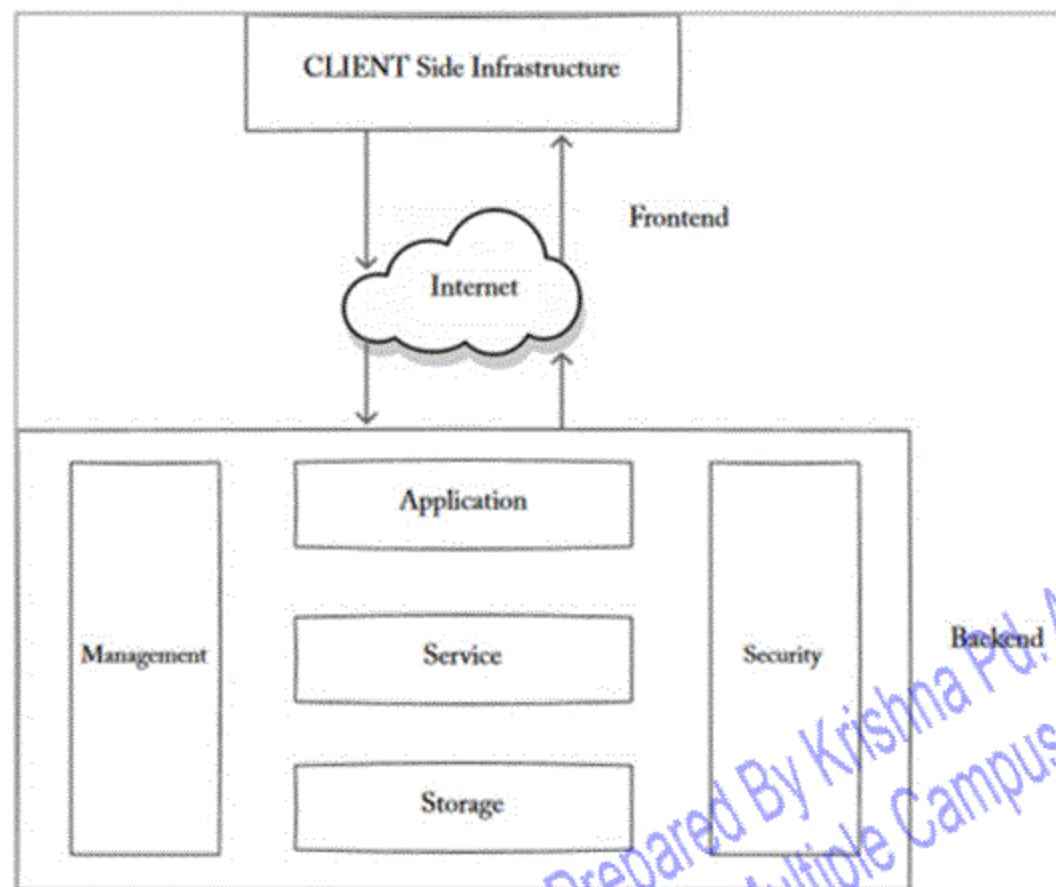
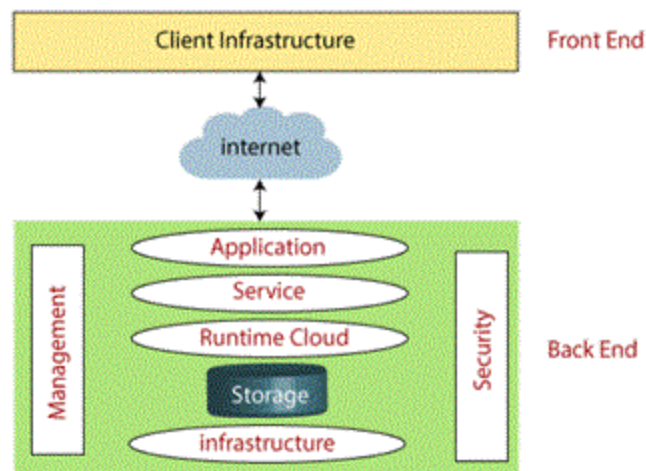
- » Cloud computing reference architecture, which identifies the **major actors, their activities and functions** in cloud computing.
- » The diagram depicts a **generic high-level architecture** and is intended to facilitate the understanding of the requirements, uses, characteristics and standards of cloud computing.
- » Cloud computing reference architecture defines **five major actors: cloud consumer, cloud provider, cloud carrier, cloud auditor and cloud broker**. Each actor is an entity (a person or an organization) that participates in a transaction or process and/or performs tasks in cloud computing. Table 1 briefly lists the actors defined in the NIST cloud computing reference architecture.

Actor	Definition
Cloud Consumer	A person or organization that maintains a business relationship with, and uses service from, <i>Cloud Providers</i> .
Cloud Provider	A person, organization, or entity responsible for making a service available to interested parties.
Cloud Auditor	A party that can conduct independent assessment of cloud services, information system operations, performance and security of the cloud implementation.
Cloud Broker	An entity that manages the use, performance and delivery of cloud services, and negotiates relationships between <i>Cloud Providers</i> and <i>Cloud Consumers</i> .
Cloud Carrier	An intermediary that provides connectivity and transport of cloud services from <i>Cloud Providers</i> to <i>Cloud Consumers</i> .

Cloud computing architecture

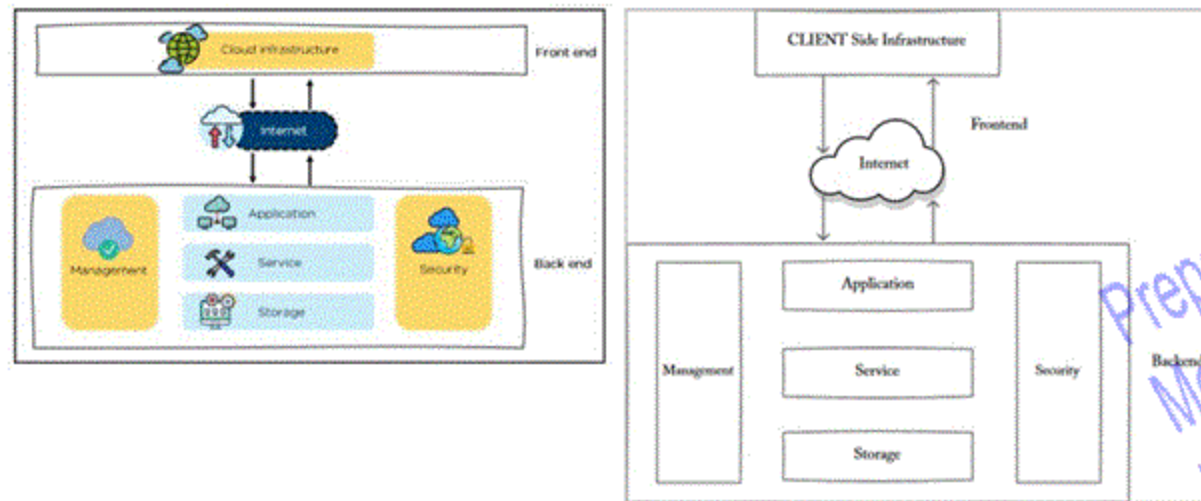


Architecture of Cloud Computing



Cloud computing architecture

- » The cloud computing architecture comprises **two fundamental components, i.e. frontend and backend**. Frontend works as a client in such **architecture and communicates** with the **backend via a network or internet**.
- » In the cloud computing architecture, the client-side or frontend is visible to the end-user. The frontend **sends queries to the backend via the middleware**.
- » A cloud computing architecture provides higher bandwidth to its users due to which data over the cloud can be used from anywhere across the world at any time. Due to its architecture, it not only shares resources among client source consumers but also with **open-source communities** like Microsoft and Red hat.
- » The **backend protects the data and respond** to the queries asked by the frontend. The backend is a bigger part of the whole cloud computing architecture as shown below:



Cloud computing architecture

Front-End

- » It provides **applications and the interfaces** that are required for the cloud-based service.
- » It consists of client's side applications, which are **web browsers** such as Google Chrome and Internet Explorer.
- » Cloud infrastructure is the only component of the front-end.
- » Cloud infrastructure consists of hardware and software **components such as data storage, server, virtualization software**, etc.
- » It also provides a **Graphical User Interface** to the end-users to perform respective tasks.

Back-End

- » It is responsible for **monitoring all the programs that run the application** on the front-end.
- » It has a large **number of data storage systems and servers**.
- » The back-end is an important and huge part of the whole cloud computing architecture, as shown above: The components of the back-end cloud architecture are mentioned next. Let's understand them in detail one by one.

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Cloud computing architecture

Application

- » It can either be a **software or a platform**
- » Depending upon the client requirement, the application **provides the result to the end user** (with resources) in the back end service.

Services

- » Service is an **essential component** in cloud architecture
- » Its responsibility is to **provide utility** in the architecture
- » In a Cloud, few widely used services among the end-users are **storage application development environments and web services**.

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Cloud computing architecture

Storage

- » It stores and maintains **data like files, videos, documents**, etc. over the internet
- » Some of the popular examples of storage services are below:
- » Amazon S3
- » Oracle Cloud-Storage
- » Microsoft Azure Storage
- » Its **capacity varies** depending upon the service providers available in the market

Management

- » Its task is to allot **specific resources to a specific task**, it simultaneously performs various functions of the cloud environment.
- » It helps in the management of components like **application, task, service, security, data storage, and cloud infrastructure**
- » In simple terms, it establishes coordination among the cloud resources.

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Infrastructure:

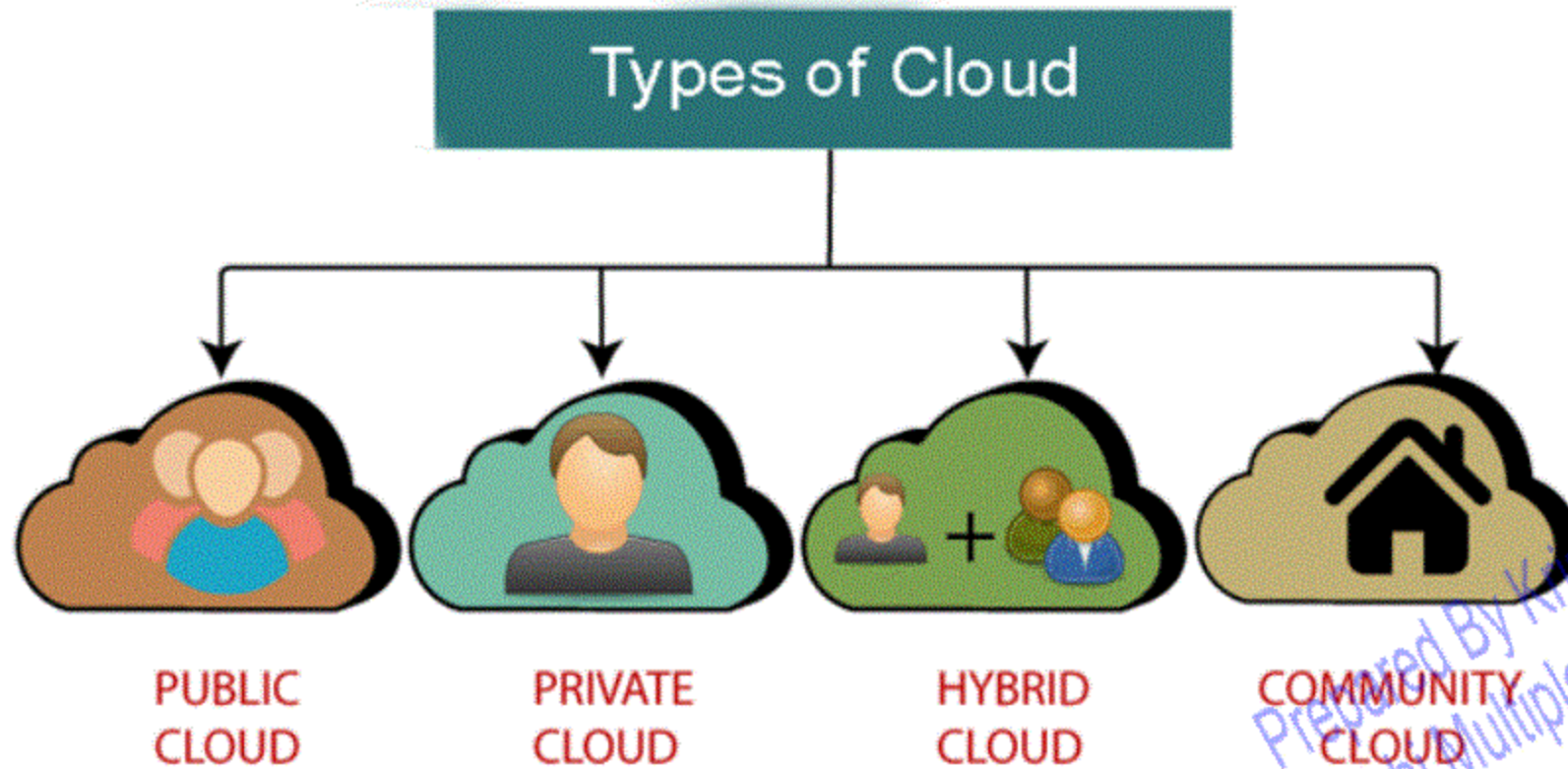
- » The engine that **steers all the cloud software services** is called infrastructure. It **includes CPU, Motherboard, Graphics Processing Unit (GPU), network cards, accelerator cards**, etc. The infrastructure models always depend on the workloads of the clients.

Security:

- » Security is an **integral and critical part** of any cloud computing infrastructure. We create **security infrastructure** by keeping the **debugging process** in mind. In case of any issue, debugging should be easy. **Regular storage backup** is the first step to ensure security in a cloud computing system. And **virtual firewalls** are other crucial elements of the cloud security infrastructure.

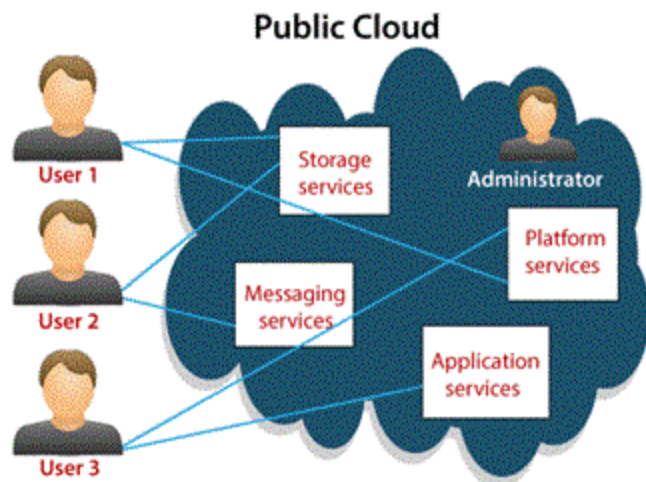
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Cloud Computing Deployment Models



Public Cloud

- » Public Cloud provides a **shared platform** that is accessible to the **general public** through an Internet connection.
- » Public cloud operated on the **pay-as-per-use model** and administrated by the **third party**, i.e., Cloud service provider.
- » In the Public cloud, the same storage is being used by multiple users at the same time.
- » Public cloud is **owned, managed, and operated** by businesses, universities, government organizations, or a combination of them.
- » Amazon Elastic Compute Cloud (EC2), Microsoft Azure, IBM's Blue Cloud, Sun Cloud, and Google Cloud are examples of the public cloud.



Advantages of Public Cloud

- » **Low Cost:** Public cloud has a lower cost than private, or hybrid cloud, as it shares the same resources with a large number of consumers.
- » **Location Independent:** Public cloud is location independent because its services are offered through the internet.
- » **Save Time:** In Public cloud, the cloud service provider is responsible for the manage and maintain data centers in which data is stored, so the cloud user can save their time to establish connectivity, deploying new products, release product updates, configure, and assemble servers.
- » **Quickly and easily set up:** Organizations can easily buy public cloud on the internet and deployed and configured it remotely through the cloud service provider within a few hours.
- » **Business Agility:** Public cloud provides an ability to elastically re-size computer resources based on the organization's requirements.
- » **Scalability and reliability:** Public cloud offers scalable (easy to add and remove) and reliable (24*7 available) services to the users at an affordable cost.

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Disadvantages of Public Cloud

- » **Low Security:** Public Cloud is less secure because resources are shared publicly.
- » **Performance:** In the public cloud, performance depends upon the speed of internet connectivity.
- » **Less customizable:** Public cloud is less customizable than the private cloud.

Private Cloud

- » Private cloud is also known as an internal cloud or corporate cloud.
- » Private cloud provides computing services to a private internal network (within the organization) and selected users instead of the general public.
- » Private cloud provides a high level of security and privacy to data through firewalls and internal hosting. It also ensures that operational and sensitive data are not accessible to third-party providers.
- » HP Data Centers, Microsoft, Elasta-private cloud, and Ubuntu are the example of a private cloud.



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Advantages and Disadvantages of Private Cloud

Advantages of Private cloud

- » **More Control:** Private clouds have more control over their resources and hardware than public clouds because it is only accessed by selected users.
- » **Security & privacy:** Security & privacy are one of the big advantages of cloud computing. Private cloud improved the security level as compared to the public cloud.
- » **Improved performance:** Private cloud offers better performance with improved speed and space capacity.

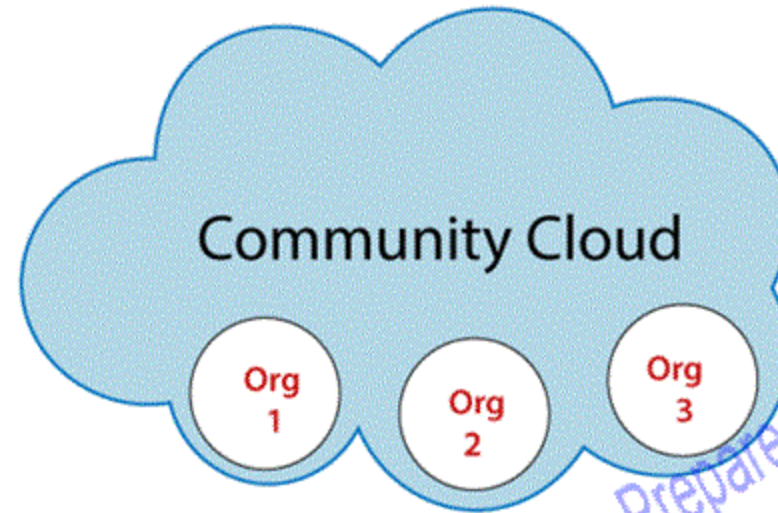
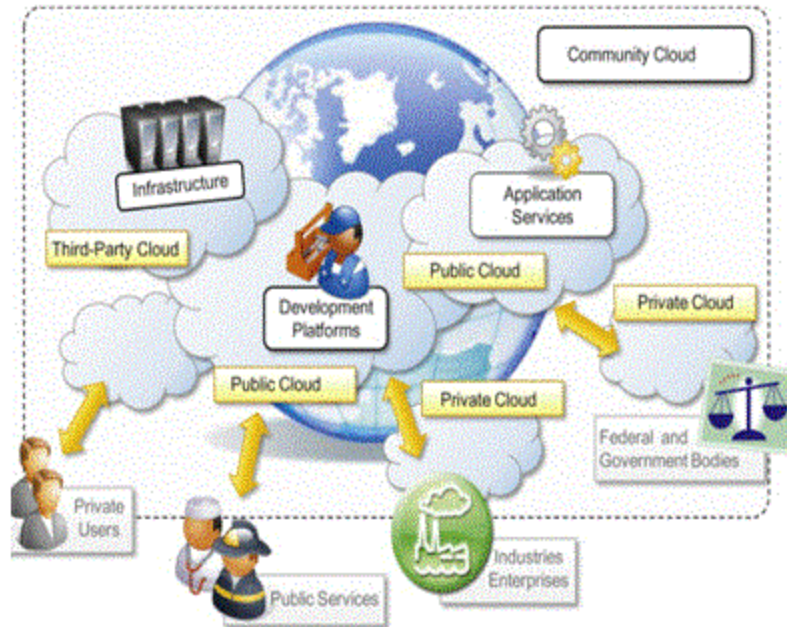
Disadvantages of Private Cloud

- » **High cost:** The cost is higher than a public cloud because set up and maintain hardware resources are costly.
- » **Restricted area of operations:** As we know, private cloud is accessible within the organization, so the area of operations is limited.
- » **Limited scalability:** Private clouds are scaled only within the capacity of internal hosted resources.
- » **Skilled people :** Skilled people are required to manage and operate cloud services.

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Community Cloud

- » **Community Cloud:** Community cloud is a cloud infrastructure that allows systems and services to be accessible by a group of several organizations to share the information.
- » It is owned, managed, and operated by one or more organizations in the community, a third party, or a combination of them.
- » **Example:** Our government organization within Nepal may share computing infrastructure in the cloud to manage data.



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Advantages of Community Cloud and Disadvantages of Community Cloud

Advantages of Community Cloud

- » **Cost effective:** Community cloud is cost effective because the whole cloud is shared between several organizations or a community.
- » **Flexible and Scalable:** The community cloud is flexible and scalable because it is compatible with every user. It allows the users to modify the documents as per their needs and requirement.
- » **Security:** Community cloud is more secure than the public cloud but less secure than the private cloud.
- » **Sharing infrastructure:** Community cloud allows us to share cloud resources, infrastructure, and other capabilities among various organizations.

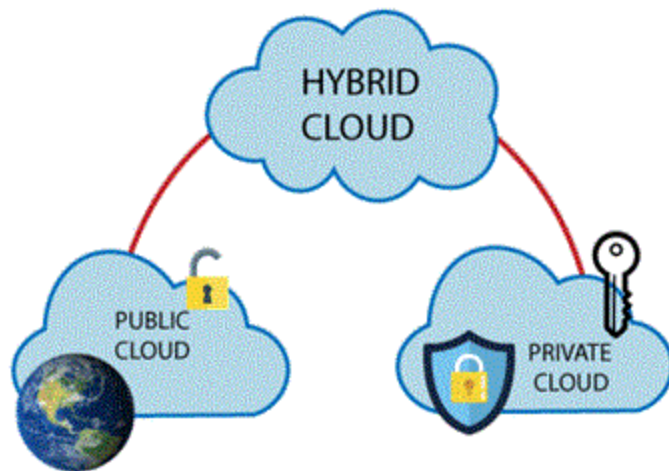
Disadvantages of Community Cloud

- » Community cloud is not a good choice for every organization.
- » Slow adoption to data
- » The fixed amount of data storage and bandwidth is shared among all community members.
- » Community Cloud is costly than the public cloud.
- » Sharing responsibilities among organizations is difficult.
- »

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Hybrid Cloud

- » Hybrid cloud is a combination of public and private clouds.
- » Hybrid cloud = public cloud + private cloud
- » The main aim to combine these cloud (Public and Private) is to create a unified, automated, and well-managed computing environment.
- » In the Hybrid cloud, non-critical activities are performed by the public cloud and critical activities are performed by the private cloud.
- » Mainly, a hybrid cloud is used in finance, healthcare, and Universities.
- » The best hybrid cloud provider companies are Amazon, Microsoft, Google, Cisco, and NetApp.



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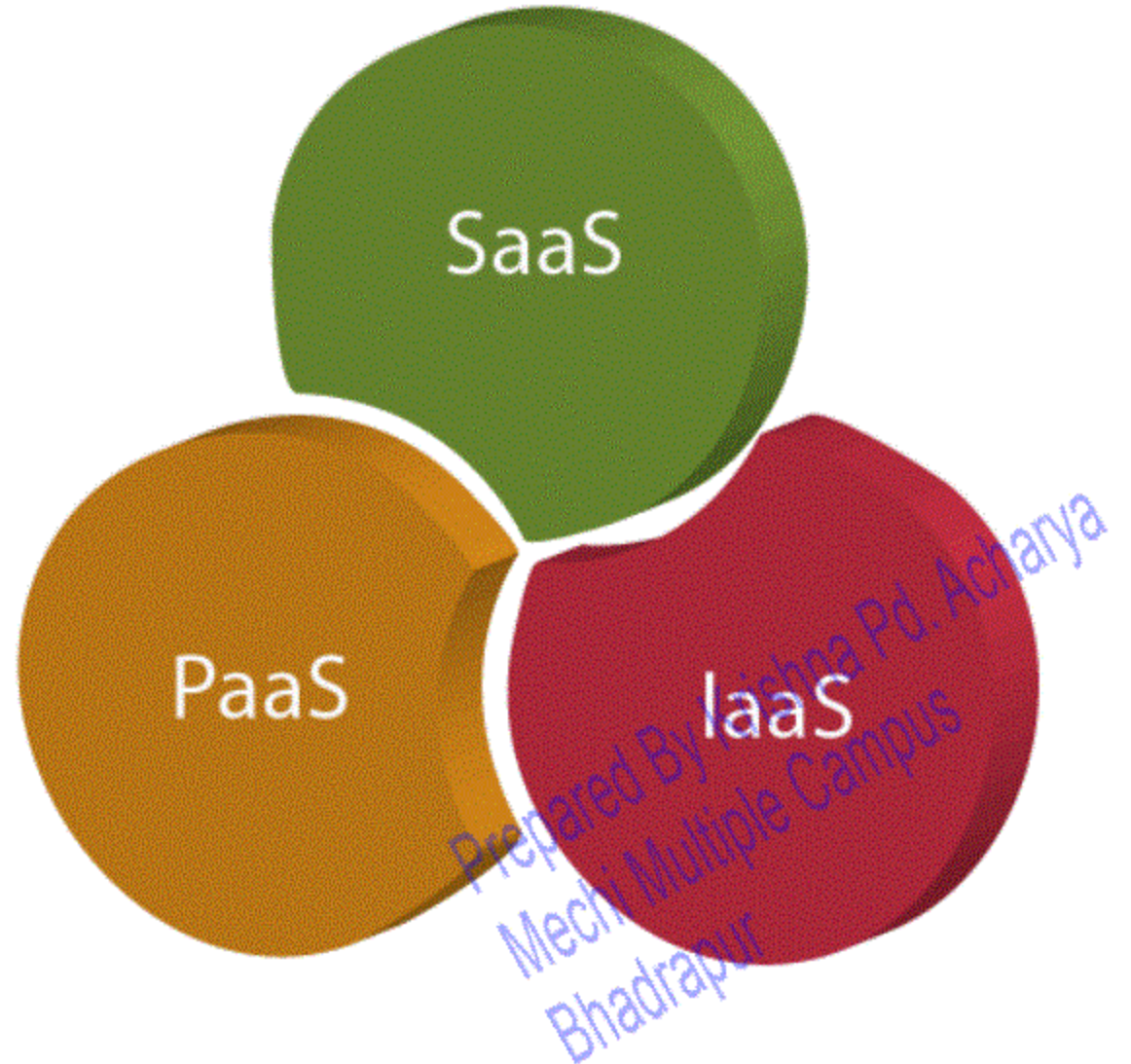
Advantages and Disadvantages of Hybrid Cloud

- » **Flexible and secure:** It provides flexible resources because of the public cloud and secure resources because of the private cloud.
- » **Cost effective:** Hybrid cloud costs less than the private cloud. It helps organizations to save costs for both infrastructure and application support.
- » **Cost effective:** It offers the features of both the public as well as the private cloud. A hybrid cloud is capable of adapting to the demands that each company needs for space, memory, and system.
- » **Security:** Hybrid cloud is secure because critical activities are performed by the private cloud.
- » **Risk Management:** Hybrid cloud provides an excellent way for companies to manage the risk.
- » **Disadvantages of Hybrid Cloud**
- » **Networking issues:** In the Hybrid Cloud, networking becomes complex because of the private and the public cloud.
- » **Infrastructure Compatibility:** Infrastructure compatibility is the major issue in a hybrid cloud. With dual-levels of infrastructure, a private cloud controls the company, and a public cloud does not, so there is a possibility that they are running in separate stacks.
- » **Reliability:** The reliability of the services depends on cloud service providers.

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Cloud Service Models

» Cloud Service Models



Infrastructure as a Service (IaaS)

- » IaaS is also known as **Hardware as a Service (HaaS)**. It is a computing infrastructure managed over the internet. The main advantage of using IaaS is that it helps users to avoid the cost and complexity of purchasing and managing the physical servers.

Characteristics of IaaS

- » Resources are available as a service
- » Services are highly scalable
- » Dynamic and flexible
- » GUI and API-based access
- » Automated administrative tasks
- » **Example:** DigitalOcean, Linode, Amazon Web Services (AWS), Microsoft Azure, Google Compute Engine (GCE), Rackspace, and Cisco Metacloud.

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Infrastructure as a Service (IaaS)

- » IaaS provider provides the following services -
- » **Compute:** Computing as a Service includes virtual central processing units and virtual main memory for the Vms that is provisioned to the end- users.
- » **Storage:** IaaS provider provides back-end storage for storing files.
- » **Network:** Network as a Service (NaaS) provides networking components such as routers, switches, and bridges for the Vms.
- » **Load balancers:** It provides load balancing capability at the infrastructure la



Advantages and disadvantages of IaaS

- » **Shared infrastructure:** IaaS allows multiple users to share the same physical infrastructure.
- » **Web access to the resources:** IaaS allows IT users to access resources over the internet.
- » **Pay-as-per-use model:** IaaS providers provide services based on the pay-as-per-use basis. The users are required to pay for what they have used.
- » **Focus on the core business:** IaaS providers focus on the organization's core business rather than on IT infrastructure.
- » **On-demand scalability:** On-demand scalability is one of the biggest advantages of IaaS. Using IaaS, users do not worry about to upgrade software and troubleshoot the issues related to hardware components.

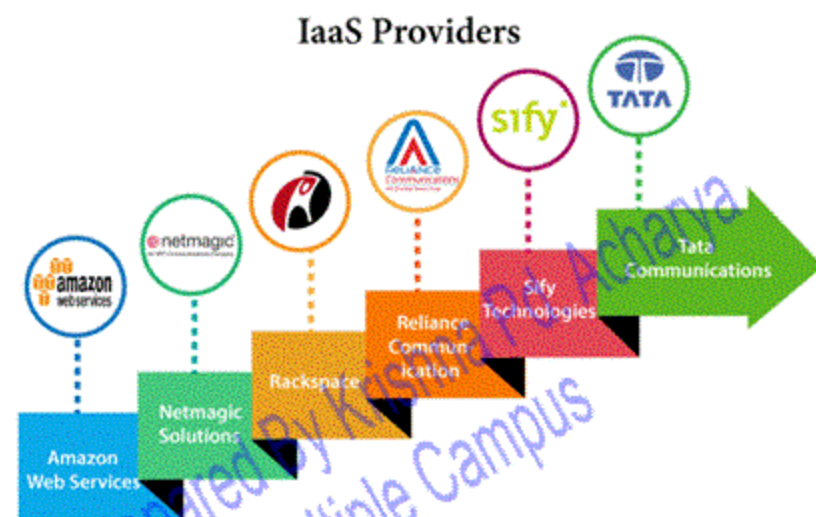
Disadvantages of IaaS cloud

- » **Security:** Security is one of the biggest issues in IaaS. Most of the IaaS providers are not able to provide 100% security.
- » **Maintenance & Upgrade :** Although IaaS service providers maintain the software, but they do not upgrade the software for some organizations.
- » **Interoperability issues:** It is difficult to migrate VM from one IaaS provider to the other, so the customers might face problem related to vendor lock-in

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IaaS

IaaS Vendor	IaaS Solution	Details
Amazon Services	Web Elastic, Elastic Compute Cloud (EC2) MapReduce, Route 53, Virtual Private Cloud, etc.	The cloud computing platform pioneer, Amazon offers auto scaling, cloud monitoring, and load balancing features as part of its portfolio.
Netmagic Solutions	Netmagic IaaS Cloud	Netmagic runs from data centers in Mumbai, Chennai, and Bangalore, and a virtual data center in the United States. Plans are underway to extend services to West Asia.
Rackspace	Cloud servers, cloud files, cloud sites, etc.	The cloud computing platform vendor focuses primarily on enterprise-level hosting services.
Reliance Communications	Reliance Internet Data Center	RIDC supports both traditional hosting and cloud services, with data centers in Mumbai, Bangalore, Hyderabad, and Chennai. The cloud services offered by RIDC include IaaS and SaaS.
Sify Technologies	Sify IaaS	Sify's cloud computing platform is powered by HP's converged infrastructure. The vendor offers all three types of cloud services: IaaS, PaaS, and SaaS.
Tata Communications	InstaCompute	InstaCompute is Tata Communications' IaaS offering. InstaCompute data centers are located in Hyderabad and Singapore, with operations in both countries.



Platform as a Service (PaaS)

- » Platform as a Service (PaaS) provides a runtime environment. It allows programmers to easily create, test, run, and deploy web applications.
- » You can purchase these applications from a cloud service provider on a pay-as-per use basis and access them using the Internet connection.
- » In PaaS, back end scalability is managed by the cloud service provider, so end- users do not need to worry about managing the infrastructure.
- » **Characteristics of PaaS**
- » Accessible to various users via the same development application.
- » Integrates with web services and databases.
- » Builds on virtualization technology, so resources can easily be scaled up or down as per the organization's need.
- » Support multiple languages and frameworks.
- » Provides an ability to "**Auto-scale**".
- » **Example:** AWS Elastic Beanstalk, Windows Azure, Heroku, Force.com, Google App Engine, Apache Stratos, Magento Commerce Cloud, and OpenShift.

Platform as a Service (PaaS)

- » **Programming languages:** PaaS providers provide various programming languages for the developers to develop the applications. Some popular programming languages provided by PaaS providers are Java, PHP, Ruby, Perl, and Go.
- » **Application frameworks:** PaaS providers provide application frameworks to easily understand the application development. Some popular application frameworks provided by PaaS providers are Node.js, Drupal, Joomla, WordPress, Spring, Play, Rack, and Zend.
- » **Databases:** PaaS providers provide various databases such as ClearDB, PostgreSQL, MongoDB, and Redis to communicate with the applications.
- » **Other tools:** PaaS providers provide various other tools that are required to develop, test, and deploy the applications.



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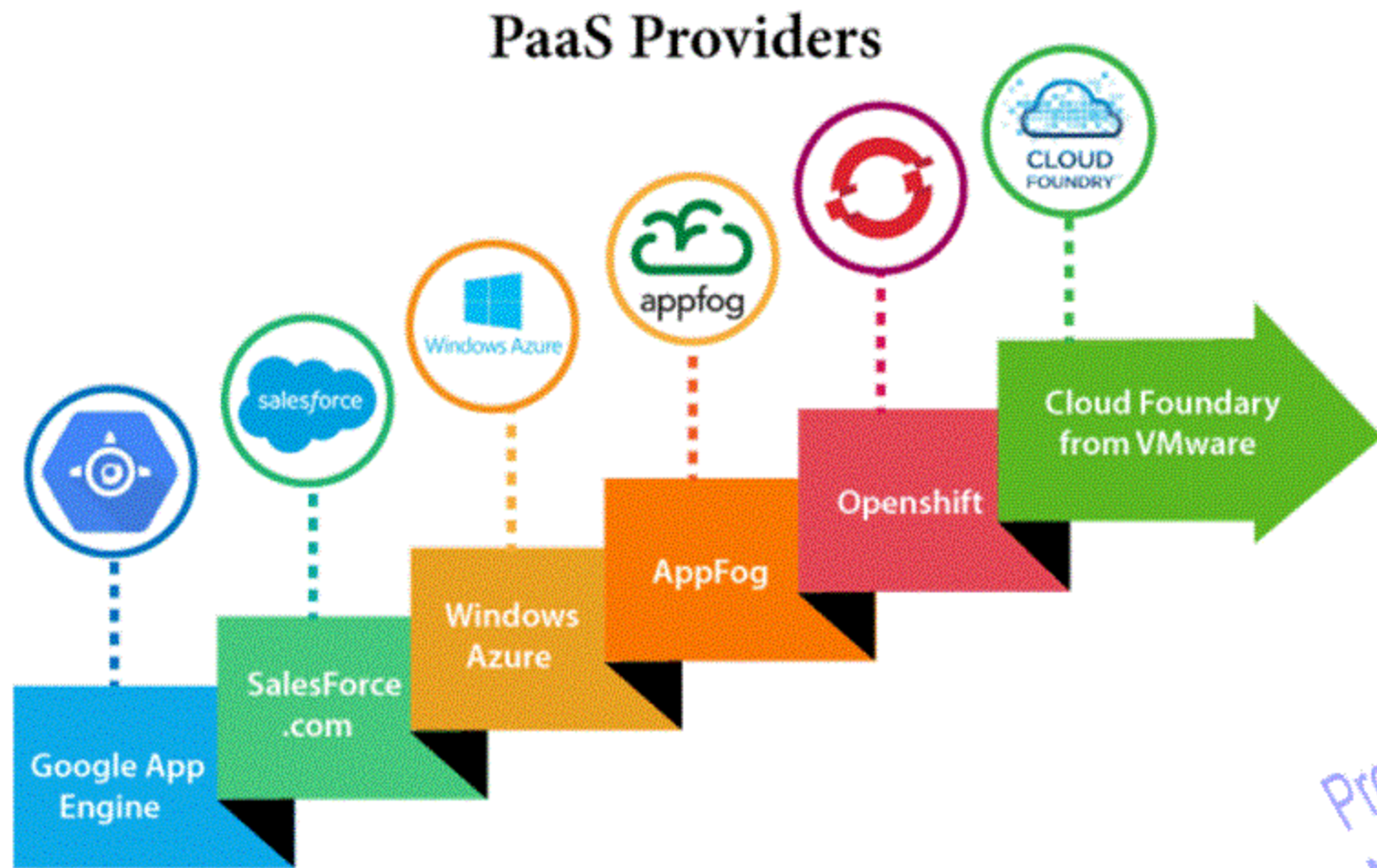
Advantages and Disadvantages of PaaS

- » **Simplified Development:** PaaS allows developers to focus on development and innovation without worrying about infrastructure management.
- » **Lower risk:** No need for up-front investment in hardware and software. Developers only need a PC and an internet connection to start building applications.
- » **Prebuilt business functionality:** Some PaaS vendors also provide already defined business functionality so that users can avoid building everything from very scratch and hence can directly start the projects only.
- » **Instant community:** PaaS vendors frequently provide online communities where the developer can get the ideas to share experiences and seek advice from others.
- » **Scalability:** Applications deployed can scale from one to thousands of users without any changes to the applications.

Disadvantages

- » **Vendor lock-in:** One has to write the applications according to the platform provided by the PaaS vendor, so the migration of an application to another PaaS vendor would be a problem.
- » **Data Privacy:** Corporate data, whether it can be critical or not, will be private, so if it is not located within the walls of the company, there can be a risk in terms of privacy of data.
- » **Integration with the rest of the systems applications:** It may happen that some applications are local, and some are in the cloud. So there will be chances of increased complexity when we want to use data which in the cloud with the local data.

PaaS Services Provider



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PaaS Services Provider

Providers	Services
Google App Engine (GAE)	App Identity, URL Fetch, Cloud storage client library, Logservice
Salesforce.com	Faster implementation, Rapid scalability, CRM Services, Sales cloud, Mobile connectivity, Chatter.
Windows Azure	Compute, security, IoT, Data Storage.
AppFog	Justcloud.com, SkyDrive, GoogleDocs
Openshift	RedHat, Microsoft Azure.
Cloud Foundry from VMware	Data, Messaging, and other services.

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Software as a Service (SaaS)

- » SaaS is also known as "**On-Demand Software**". It is a software distribution model in which services are hosted by a cloud service provider. These services are available to end-users over the internet so, the end-users do not need to install any software on their devices to access these services.
- » There are the following services provided by SaaS providers -
- » **Business Services** - SaaS Provider provides various business services to start-up the business. The SaaS business services include **ERP** (Enterprise Resource Planning), **CRM** (Customer Relationship Management), **billing**, and **sales**.
- » **Document Management** - SaaS document management is a software application offered by a third party (SaaS providers) to create, manage, and track electronic documents.
- » **Example:** Slack, Samepage, Box, and Zoho Forms.
- » **Social Networks** - As we all know, social networking sites are used by the general public, so social networking service providers use SaaS for their convenience and handle the general public's information.
- » **Mail Services** - To handle the unpredictable number of users and load on e-mail services, many e-mail providers offering their services using SaaS.



Advantages of SaaS cloud

- » **SaaS is easy to buy:** SaaS pricing is based on a monthly fee or annual fee subscription, so it allows organizations to access business functionality at a low cost, which is less than licensed applications.
- » Unlike traditional software, which is sold as a licensed based with an up-front cost (and often an optional ongoing support fee), SaaS providers are generally pricing the applications using a subscription fee, most commonly a monthly or annually fee.
- » **One to Many:** SaaS services are offered as a one-to-many model means a single instance of the application is shared by multiple users.
- » **Less hardware required for SaaS:** The software is hosted remotely, so organizations do not need to invest in additional hardware.
- » **Low maintenance required for SaaS:** Software as a service removes the need for installation, set-up, and daily maintenance for the organizations. The initial set-up cost for SaaS is typically less than the enterprise software.
- » SaaS vendors are pricing their applications based on some usage parameters, such as a number of users using the application. So SaaS does easy to monitor and automatic updates.
- » **No special software or hardware versions required:** All users will have the same version of the software and typically access it through the web browser. SaaS reduces IT support costs by outsourcing hardware and software maintenance and support to the IaaS provider.
- » **Multidevice support:** SaaS services can be accessed from any device such as desktops, laptops, tablets, phones, and thin clients.
- » **API Integration:** SaaS services easily integrate with other software or services through standard APIs.
- » **No client-side installation:** SaaS services are accessed directly from the service provider using the internet connection, so do not need to require any software installation.

Disadvantages of SaaS cloud

- » **Security:** Actually, data is stored in the cloud, so security may be an issue for some users. However, cloud computing is not more secure than in-house deployment.
- » **Latency issue:** Since data and applications are stored in the cloud at a variable distance from the end-user, there is a possibility that there may be greater latency when interacting with the application compared to local deployment. Therefore, the SaaS model is not suitable for applications whose demand response time is in milliseconds.
- » **Total Dependency on Internet:** Without an internet connection, most SaaS applications are not usable.
- » **Switching between SaaS vendors is difficult :** Switching SaaS vendors involves the difficult and slow task of transferring the very large data files over the internet and then converting and importing them into another SaaS also.

Popular SaaS Providers



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Popular SaaS Providers

Provider		Services
Salseforce.com		On-demand CRM solutions
Microsoft Office 365		Online office suite
Google Apps		Gmail, Google Calendar, Docs, and sites
NetSuite		ERP, accounting, order management, CRM, Professionals Services Automation (PSA), and e-commerce applications.
GoToMeeting		Online meeting and video-conferencing software
Constant Contact		E-mail marketing, online survey, and event marketing
Oracle CRM		CRM applications
Workday, Inc		Human capital management, payroll, and financial management.

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Benefits of Cloud Computing Architecture

- » The cloud computing architecture is designed in such a way that:
- » It solves latency issues and improves data processing requirements
- » It reduces IT operating costs and gives good accessibility to access data and digital tools
- » It helps businesses to easily scale up and scale down their cloud resources
- » It has a flexibility feature which gives businesses a competitive advantage
- » It results in better disaster recovery and provides high security
- » It automatically updates its services
- » It encourages remote working and promotes team collaboration

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SOA (Service Oriented Architecture)

- » A Service-Oriented Architecture or SOA is a design pattern which is designed to build distributed systems that deliver services to other applications through the protocol. It is only a concept and not limited to any programming language or platform.
- » Service-oriented architecture (SOA) is an type of architecture used in software design that supports service orientation, whereby services are provided to other components through a communication protocol over a network.
- » Service-oriented architecture (SOA) is a style of software design where services are provided to the other components by application components, through a communication protocol over a network.
- » The basic principles of service-oriented architecture are independent of vendors, products and technologies. A service is a discrete unit of functionality that can be accessed remotely and acted upon and updated independently, such as retrieving a credit card statement online.
- » A service has four properties according to one of many definitions of SOA:
 - » It logically represents a business activity with a specified outcome.
 - » It is self-contained.
 - » It is a black box for its consumers.
 - » It may consist of other underlying services.

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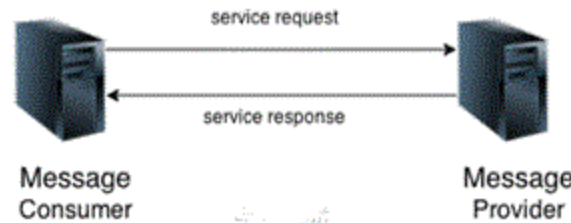
There are two major roles within Service-oriented Architecture

Service

- » A service is a well-defined, self-contained function that represents a unit of functionality. A service can exchange information from another service. It is not dependent on the state of another service. It uses a loosely coupled, message-based communication model to communicate with applications and other services.

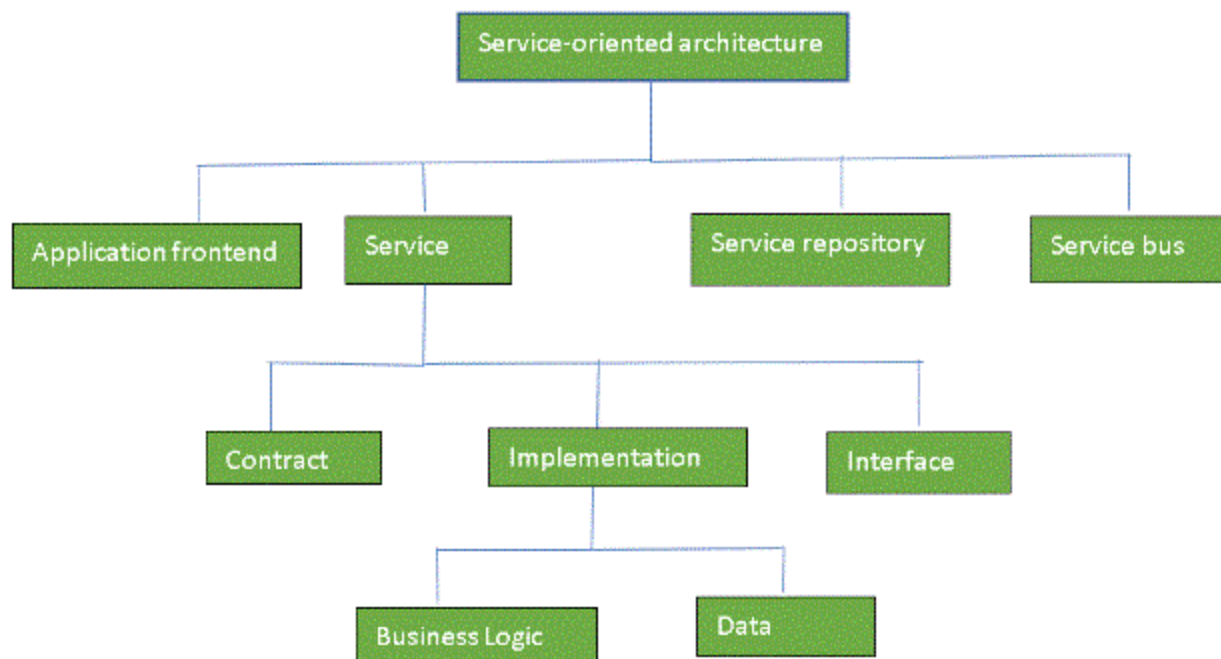
Service Connections

- » The figure given below illustrates the service-oriented architecture. Service consumer sends a service request to the service provider, and the service provider sends the service response to the service consumer. The service connection is understandable to both the service consumer and service provider.



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Components of SOA



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SOA (Service Oriented Architecture): Seven SOA steps

- » Create/Expose Services
- » Register Services
- » Secure Services
- » Manage (monitor) Services
- » Mediate and Virtualize Services
- » Govern the SOA
- » SOA Integration With ESB(Enterprise Services Bus)

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Components of SOA

Guiding Principles of SOA:

- » **Standardized service contract:** Specified through one or more service description documents.
- » **Loose coupling:** Services are designed as self-contained components, maintain relationships that minimize dependencies on other services.
- » **Abstraction:** A service is completely defined by service contracts and description documents. They hide their logic, which is encapsulated within their implementation.
- » **Reusability:** Designed as components, services can be reused more effectively, thus reducing development time and the associated costs.
- » **Autonomy:** Services have control over the logic they encapsulate and, from a service consumer point of view, there is no need to know about their implementation.
- » **Discoverability:** Services are defined by description documents that constitute supplemental metadata through which they can be effectively discovered. Service discovery provides an effective means for utilizing third-party resources.
- » **Composability:** Using services as building blocks, sophisticated and complex operations can be implemented. Service orchestration and choreography provide a solid support for composing services and achieving business goals.

Advantages and Disadvantages of SOA

Service reusability: In SOA, applications are made from existing services. Thus, services can be reused to make many applications.

Easy maintenance: As services are independent of each other they can be updated and modified easily without affecting other services.

Platform independent: SOA allows making a complex application by combining services picked from different sources, independent of the platform.

Availability: SOA facilities are easily available to anyone on request.

Reliability: SOA applications are more reliable because it is easy to debug small services rather than huge codes.

Scalability: Services can run on different servers within an environment, this increases scalability.

Disadvantages of SOA:

- » **High overhead:** A validation of input parameters of services is done whenever services interact, this decreases performance as it increases load and response time.
- » **High investment:** A huge initial investment is required for SOA.
- » **Complex service management:** When services interact they exchange messages to tasks. The number of messages may go in millions. It becomes a cumbersome task to handle a large number of messages.

Security, trust, and privacy Security

- » Security, trust, and privacy Security, issues are major obstacles for massive adoption of cloud computing. The traditional cryptographic technologies are used to prevent data tampering and access to sensitive information. The massive use of virtualization technologies exposes the existing system to new threats, which previously were not considered applicable.
- » For example, it might be possible that applications hosted in the cloud can process sensitive information; such information can be stored within a cloud storage facility using the most advanced technology in cryptography to protect data and then be considered safe from any attempt to access it without the required permissions.
- » Although these data are processed in memory, they must necessarily be decrypted by the legitimate application, but since the application is hosted in a managed virtual environment it becomes accessible to the virtual machine manager that by program is designed to access the memory pages of such an application.
- » In this case, what is experienced is a lack of control over the environment in which the application is executed, which is made possible by leveraging the cloud. It then happens that a new way of using existing technologies creates new opportunities for additional threats to the security of applications.
- » The lack of control over their own data and processes also poses severe problems for the trust we give to the cloud service provider and the level of privacy we want to have for our data.

Security, trust, and privacy Security

- » On one side we need to decide whether to trust the provider itself; on the other side, specific regulations can simply prevail over the agreement the provider is willing to establish with us concerning the privacy of the information managed on our behalf.
- » Moreover, cloud services delivered to the end user can be the result of a complex stack of services that are obtained by third parties via the primary cloud service provider. In this case there is a chain of responsibilities in terms of service delivery that can introduce more vulnerability for the secure management of data, the enforcement of privacy rules, and the trust given to the service provider.
- » In particular, when a violation of privacy or illegal access to sensitive information is detected, it could become difficult to identify who is liable for such violations. The challenges in this area are, then, mostly concerned with devising secure and trustable systems from different perspectives: technical, social, and legal.

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Application

- » SOA is used in many ways around us whether it is mentioned or not.
- » SOA infrastructure is used by many armies and air forces to deploy situational awareness systems.
- » SOA is used to improve healthcare delivery.
- » Nowadays many apps are games and they use inbuilt functions to run. For example, an app might need GPS so it uses the inbuilt GPS functions of the device. This is SOA in mobile solutions.
- » SOA helps maintain museums a virtualized storage pool for their information and content.

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Assignment

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